A Laminate Parallel Articulated Spine for a Cat-Scale Quadruped Robot

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Abstract

This project presents the design and physical prototyping of a bio-inspired mechanism intended to serve as the articulated spine of a small quadruped robot with Jansen mechanisms for its four legs. Taking inspiration from the classical Sarrus linkage, we develop a planar articulated spine composed of a three-branch parallel linkage (3R–2R–3R) that connects two rigid 3D-printed chassis plates and is intended to provide vertical support while allowing controlled in-plane bending. The spine is realized in a Smart Composite Microstructures (SCM) format using a five-layer laminate (rigid–adhesive–flexure–adhesive–rigid). Stiff outer layers form the links, while the flexure layer creates passive spring-like joints that generate a restoring torque as the spine bends. The overall morphology is scaled to roughly match the proportions of a juvenile cat while operating at a much lower mass of about 250 g. We construct both a paper analogue of the mechanism and a laminate prototype integrated into a quadruped chassis. A planar kinematic model, implemented in Python, represents the 3R–2R–3R parallel spine in the sagittal plane and is used to compute representative joint torques and joint velocities for expected ground reaction forces and vertical motions of the front chassis. Using this model, we compute representative joint torques and joint velocities for expected ground reaction forces and vertical motions of the front chassis. The results demonstrate that a linkage-based, passively compliant spine is feasible within the constraints of laminate fabrication and provides a structured way to tune the directional stiffness of the robot torso for future integration with Jansen-leg quadruped locomotion.

1 Introduction

Bio-inspired legged robots often borrow structural ideas from animals to improve stability, efficiency, and maneuverability. Many existing quadruped designs use a rigid torso with compliant legs, or a uniformly flexible backbone that bends easily in all directions. While these approaches simplify design and control, they limit the role of the spine to either a passive connector or a uniformly compliant beam, which can reduce force transmission and restrict useful body motion during locomotion.

In contrast, animals such as cats and cheetahs rely on spines that are stiff in some directions and compliant in others, allowing the torso to store and release energy, extend stride length, and stabilize the body during fast gaits. Motivated by this idea, this project explores a mechanical articulated spine for a small quadruped robot driven by Jansen mechanisms at each leg. The design is inspired by the classical Sarrus linkage and realized as a planar articulated spine composed of a three-branch parallel linkage (3R–2R–3R) that connects two rigid 3D-printed chassis plates. The links are made from stiff laminate layers, while the joints are implemented as thin flexures that behave like passive spring-like hinges due to the material properties. The goal is to achieve a spine that provides vertical support, allows controlled in-plane bending, and naturally generates a restoring torque as it deflects.

This report focuses on the mechanism-level design, scaling, and physical prototyping of this spine using laminate construction. Dynamic behavior and full leg–spine coupling are left for future work.

1.1 Project Rationale and Scope

Scope. This project makes the spine the central design element of a small quadruped platform:

1. **Parallel Articulated Spine.** Design and prototype a Sarrus-inspired planar parallel linkage that connects a front and rear chassis, targeting high vertical stiffness with controlled in-plane bending between the two bodies.

2. **Five-Layer Laminate Implementation.** Fabricate the spine links and joints using a five-layer SCM stack (rigid–adhesive–flexure–adhesive–rigid) so that links remain stiff while joints behave as compliant, spring-like flexures.

3. **Geometric Compatibility with Jansen Legs.** Dimension the spine assuming that each chassis will carry a planar, single-degree-of-freedom Jansen leg mechanism that provides a more biologically relevant foot trajectory than simple rotary legs, even though full leg–spine integration is deferred to later work.

The work in this report is limited to geometric design, basic stiffness reasoning, and fabrication of spine prototypes using laminate construction with 3D-printed chassis interfaces.

Impact. By replacing a rigid or uniformly flexible torso with a parallel articulated linkage whose joints behave as passive spring-like elements, this project isolates the spine as a key morphological variable. The prototype provides initial insight into how a Sarrus-inspired, passively compliant spine can transmit loads between the front and rear chassis while still permitting useful bending and self-centering behavior. These results inform future quadruped designs that must balance support, compliance, and integration with complex leg mechanisms such as Jansen linkages.

2 Bio-Inspiration and System Scaling

2.1 Biological Basis and Hybrid Concept

The proposed robot combines two distinct bio-inspired ideas into a single hybrid morphology. At the *body-plan level*, the platform is scaled to a small quadruped, roughly corresponding to a juvenile domestic cat with four primary limbs and a compact trunk. Quadrupeds of this size use their spine to support body weight while allowing controlled bending in the sagittal plane to extend stride length and smooth center-of-mass motion during locomotion.

At the *mechanism level*, the internal backbone draws inspiration from myriapod-style robots that use Sarrus-like linkages to realize segmented, popup-compatible trunks. Prior centimeter-scale "micropod" designs employ Sarrus mechanisms and laminate flexures to create compliant, vertically supportive backbones using only revolute joints. In this project we adapt that idea to a quadruped context: two rigid 3D-printed chassis plates (front and rear) are connected by a Sarrus-inspired parallel articulated spine, implemented in a five-layer laminate stack. Stiff outer layers form the links, while the thin flexure layer creates joints that act as passive torsional springs and generate a restoring torque as the spine bends.

This results in a hybrid system: externally, the robot resembles a cat-sized quadruped with four Jansen mechanisms providing leg motion; internally, the connection between the front and rear bodies behaves like a myriapod-inspired, Sarrus-based articulated spine.

2.2 Scale Selection

The overall geometric and dynamic scale of the robot is chosen to be proportionally similar to a juvenile cat while keeping the total mass much lower to match the capabilities of laminate fabrication and small actuators. Table 1 summarizes the key parameters used for this scaling. The robot body length (distance between front and rear chassis) is set to 0.25 m, with an approximate total mass of 0.25 kg.

Table 1: Key morphological and locomotion parameters for a juvenile cat and the proposed quadruped robot. Biological values are approximate and used only for scaling.

Quantity	**Symbol**	**Juvenile cat**	**Robot target**
Body mass	m	1.5 kg to 2.0 kg	0.25 kg
Body length (shoulder–hip)	L_{body}	0.25 m to 0.30 m	0.25 m
Typical walking speed	v	0.4 m/s to 0.6 m/s	0.2 m/s to 0.3 m/s
Stride length	L_{stride}	0.30 m to 0.40 m	0.20 m to 0.25 m
Step frequency	f_{step}	1.5 Hz to 2.0 Hz	1.0 to 1.5
Max foot height (swing)	h_{foot}	0.03 m to 0.05 m	0.02 m to 0.03 m
Peak vertical GRF per leg	$F_{z,\text{max}}$	20 N to 30 N	3 N to 5 N (design)
Spine flexion range (sagittal)	θ_{spine}	$\approx 10°$	$\approx 10°$

The robot mass is intentionally much lower than that of the biological counterpart, which reduces expected ground reaction forces and laminate stresses. With a mass of 0.25 kg, the body weight is approximately 2.45 N. Designing for peak vertical ground reaction forces of 3 N to 5 N per leg provides a safety margin for dynamic effects while remaining compatible with small actuators and laminate flexures.

2.3 Supporting Literature

Several prior works inform the specific design and fabrication choices in this project.

Bing et al. [3] experimentally demonstrated that lateral spinal flexion in a small legged robot can significantly improve performance and turning behavior. Their results support the focus on spinal stiffness as a key design variable. While their spine is more uniformly compliant, this work extends the idea by investigating a directionally stiff, linkage-based spine with passive spring-like joints.

Hoover and Fearing [1] established the Smart Composite Microstructures (SCM) process for folded millirobots. This project directly adopts their five-layer laminate methodology—laser-machining rigid layers and laminating them with Kapton flexures—to realize both stiff links and compliant joints in the spine. Their analysis of laminate linkages confirms that mechanisms of similar complexity to the proposed spine (and future Jansen legs) are manufacturable at this scale.

Eckert et al. [2] compared different spine and leg designs for small quadrupeds and showed that spine and leg dynamics are tightly coupled; changes in leg trajectory often require corresponding adjustments in spinal stiffness to maintain efficiency. This motivates designing the spine with explicit consideration of its eventual interaction with Jansen legs, even though only the spine is prototyped here.

Dürr et al. [4] provided a comprehensive biomimetic foundation for legged locomotion and inter-limb coordination. While their focus is on multi-legged systems, their discussion of stability, duty factor, and phase relationships informs the long-term goal of embedding the articulated spine within a biologically grounded gait framework once the full quadruped (spine + Jansen legs) is assembled.

Novelty. Taken together, these works motivate but do not duplicate our approach. Prior robots typically either use rigid torsos with complex legs, or uniformly compliant spines with simple legs. In contrast, this project targets a cat-scale quadruped with a linkage-based, directionally stiff spine designed explicitly for integration with Jansen-style leg mechanisms. To our knowledge, a laminate Sarrus-based spine configured as the primary torso element of a quadruped has not been previously reported.

3 Robot Specifications

All key physics-based parameters for the proposed quadruped robot are collected in Table 2. Values are expressed in SI units and are drawn from biological scaling, material data, or simple calculations as indicated.

Table 2: Specifications table for the proposed quadruped robot.

Parameter	**Symbol**	**Unit**	**Value / Range**	**Sourc**
Total mass	m	kg	0.25	Measu
Body length (shoulder–hip)	L_{body}	m	0.25	Geome
Target walking speed	v	m/s	0.2–0.3	Design
Stride length	L_{stride}	m	0.20–0.25	Design
Step frequency	f_{step}	Hz	1.0–1.5	Design
Peak vertical GRF per leg	$F_{z,\max}$	N	3–5	Scaled
Body weight	$W = mg$	N	≈ 2.45	Calcul
Peak COM vertical acceleration	a_z	m/s^2	≈ 6.2	Calcul
Max spine pitch (front–rear)	$\theta_{\max}$	rad	≈ 0.17	Design
Target spine torsional stiffness	k_θ	N m/rad	≈ 1.4	Calcul
Laminate link modulus (FR4)	E_{link}	GPa	15–25	Datash
Flexure modulus (Kapton)	E_{flex}	GPa	2.5–3.0	Datash
Laminate stack	–	–	5-layer (rigid–adhesive–flexure–adhesive–rigid)	SCM p

The values in Table 2 combine biological scaling and conservative design choices. The peak ground reaction forces are scaled from biological values but slightly overestimated to provide a safety margin. The target walking speed, stride length, and step frequency are reduced relative to biology to reflect the limitations of small actuators and friction in laminate joints. The spine pitch limit and corresponding torsional stiffness are design targets chosen to allow visible bending while maintaining a strong passive restoring torque from the flexure-based joints.

4 Physical Mechanism Prototypes

4.1 Paper Mechanism Prototype

To develop intuition for the articulated spine, we first built a paper mechanism using slit-and-fold joints. Figure 1 shows the resulting model in several configurations. The prototype consists of two large prismatic frames representing the front and rear body segments and a central spine module built from folded triangular facets. All joints are implemented as thin paper folds, which act as flexible hinges.

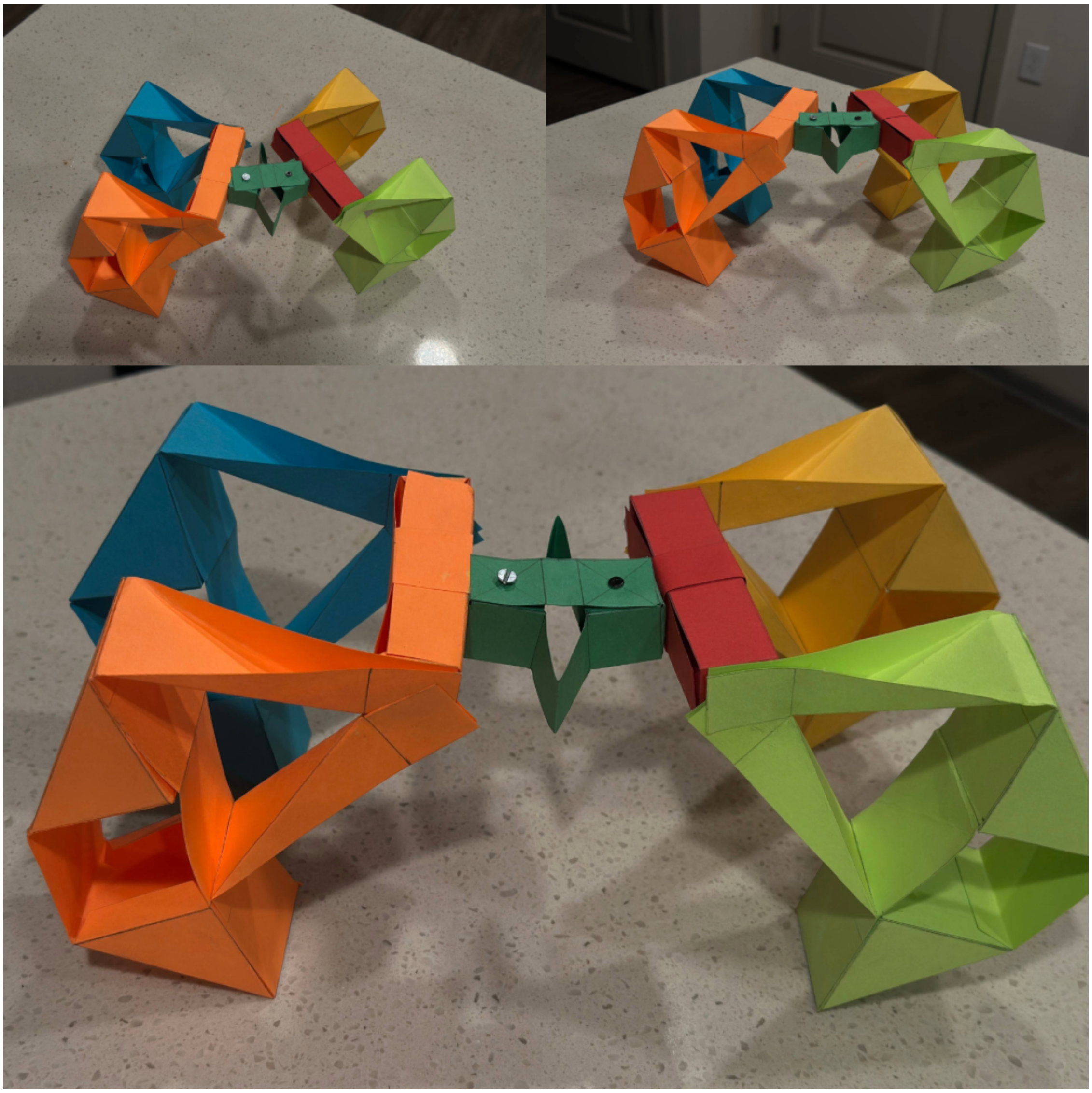

Figure 1: Paper prototype of the articulated spine mechanism in multiple configurations. The colored prismatic frames represent the front and rear body segments. The central colored elements form a spatial Sarrus-like linkage whose folded joints behave as passive hinges. By manually rotating the outer frames relative to each other, the spine bends while remaining vertically supportive, illustrating the intended behavior of the laminate spine in the final robot.

This paper analogue validates the qualitative motion of the Sarrus-based spine and helps identify locations where flexures and rigid links should be reinforced in the laminate design.

5 Mechanism Geometry and Kinematic Diagram

The physical spine module is a planar articulated spine composed of a three-branch parallel linkage (3R–2R–3R) between two rigid platforms (front and rear chassis). For clarity, we draw a planar kinematic diagram in the sagittal plane, shown in Fig. 2. The ground link of length l_0 connects points P_6 and P_0, which lie on the rear chassis. Two mirrored three-link chains connect P_6 and P_0 to a common point P_3, which represents the front chassis attachment. A central link of length l_6 connects P_6 to P_3 and enforces the parallel relationship between front and rear platforms.

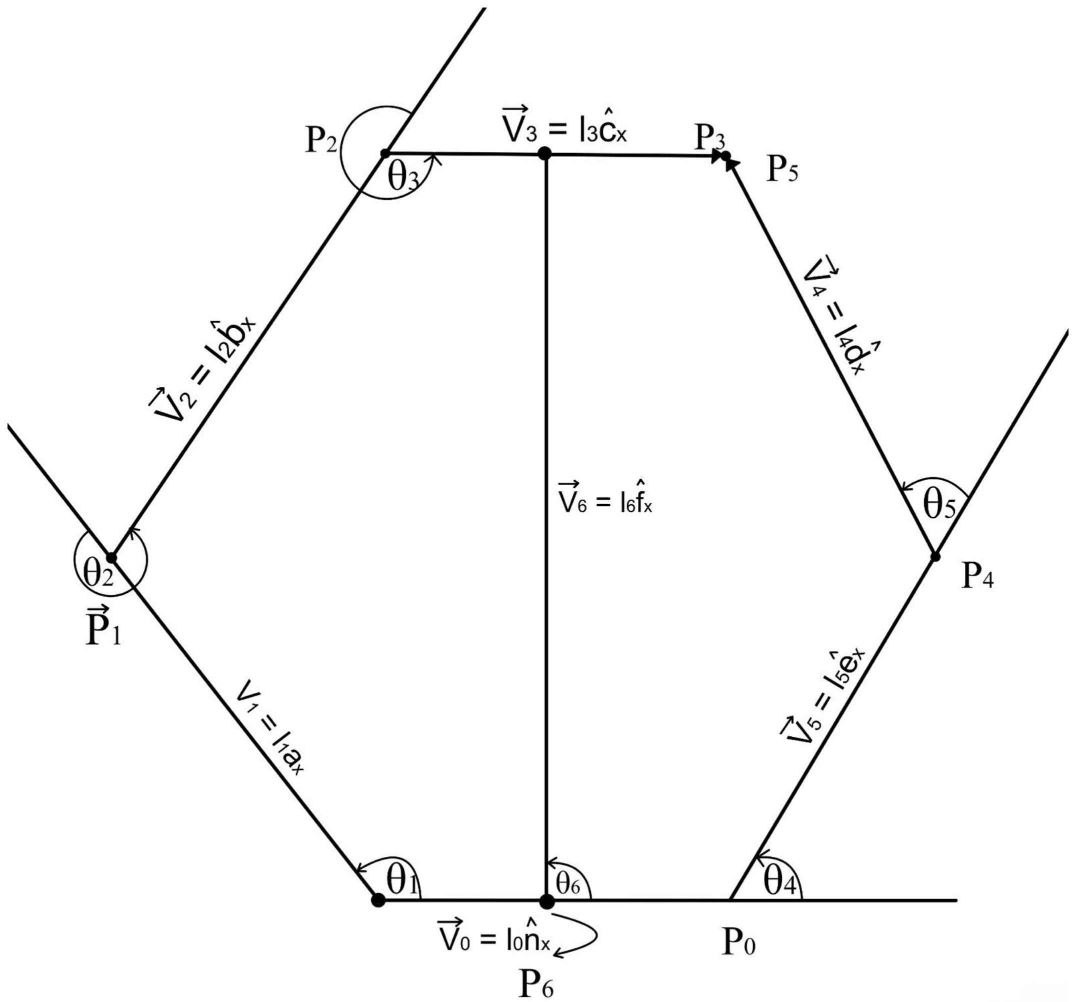

Figure 2: Planar kinematic diagram of the articulated spine. The inertial frame $\mathcal{N}$ is defined by $\{\hat{n}_x, \hat{n}_y, \hat{n}_z\}$. Rigid links are represented by vectors $\vec{V}_i = l_i\hat{(\cdot)}_x$, where l_i is the length of link i and the local $\hat{(\cdot)}_x$ axis is aligned with that link. Points P_0–P_6 denote joint locations, and joint angles $\theta_1 \ldots \theta_6$ are measured relative to the $\hat{n}_x$ axis. The full mechanism is a spatial Sarrus-based linkage; here we restrict attention to motion in the sagittal plane and model the planar 3R–2R–3R parallel spine using the joint coordinates $\theta_1 \ldots \theta_6$ defined above.

6 Estimated Goal Performance Metrics

The literature provides approximate values for body mass and ground reaction forces, but it does not directly specify several design metrics important for the articulated spine. Here we estimate (1) the peak vertical acceleration of the body during stance and (2) a target torsional stiffness for the spine.

6.1 Peak Vertical Acceleration

With total mass $m = 0.25\,\mathrm{kg}$, the robot weight is

$$W = mg = 0.25 \times 9.81 \approx 2.45\,\mathrm{N}. \tag{1}$$

During walking, each leg experiences a peak vertical ground reaction force in the range of 3–5 N. We choose a representative value $F_{z,\max} = 4\,\mathrm{N}$ for a single leg. Assuming that at the instant of peak force the net vertical force on the body is dominated by the difference between the ground reaction force and weight, Newton's second law gives

$$F_{z,\max} - W = m a_z. \tag{2}$$

Solving for a_z,

$$a_z = \frac{F_{z,\max} - W}{m} \approx \frac{4.0 - 2.45}{0.25} \approx 6.2\,\mathrm{m/s^2}, \tag{3}$$

corresponding to a peak upward acceleration of roughly $0.6g$. This sets a reasonable target for the vertical dynamics that the articulated spine must accommodate without excessive deflection.

6.2 Target Torsional Stiffness of the Spine

We would like the spine to permit a maximum relative pitch angle of $\theta_{\max} \approx 10^\circ$ between the front and rear chassis under load while generating a restoring torque that tends to re-center the body. If we assume that a disturbance produces an effective moment of $\tau_{\max} = 0.25\,\mathrm{N\,m}$ about the spine, and model the spine as a linear torsional spring,

$$\tau = k_\theta \theta, \tag{4}$$

then a target torsional stiffness is

$$k_\theta = \frac{\tau_{\max}}{\theta_{\max}} \approx \frac{0.25}{0.174} \approx 1.4\,\mathrm{N\,m/rad}. \tag{5}$$

This value serves as a goal for the combined effect of all flexure joints in the articulated spine.

6.3 Estimated Power at the Spine Joint

Combining the force and velocity estimates provides an order-of-magnitude power requirement. Using a representative vertical load of $F_y \approx 4$ N at the spine endpoint and a small vertical velocity of $\dot{y} \approx 0.05$ m/s (as considered in Sec. 7), the mechanical power transmitted through the spine is approximately

$$P \approx F_y \dot{y} \approx 4 \times 0.05 = 0.20 \text{ W}. \tag{6}$$

At the joint level, the corresponding power at joint θ_6 is $P_6 \approx \tau_6 \dot{\theta}_6$, where τ_6 and $\dot{\theta}_6$ follow from the Jacobian-based force and velocity mappings. Using typical values of $\tau_6 \sim 1$ N m and $\dot{\theta}_6 \sim 0.2$ rad/s yields $P_6 \sim 0.2$ W, consistent with the endpoint estimate and within the capabilities of small hobby servomotors.

7 Python-Based Kinematic Model

The articulated spine is modeled as a planar three-branch parallel linkage (3R–2R–3R), with geometry corresponding to the kinematic diagram in Fig. 2. A short Python script is used to evaluate the kinematics numerically and to compute Jacobians for force and velocity analysis.

7.1 Variables, Link Vectors, and Points

All motion is defined in an inertial frame $\mathcal{N}$ with basis $\{\hat{n}_x, \hat{n}_y, \hat{n}_z\}$, where $\hat{n}_x$ is horizontal and $\hat{n}_y$ is vertical. The ground origin is $O = [0, 0, 0]^T$.

The model uses:

- Link lengths $l_0, \ldots, l_6$ (constants),
- Joint angles $\theta_1, \ldots, \theta_6$ (state variables).

Planar rotations about $\hat{n}_z$ are represented by

$$R_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{7}$$

Unit vectors aligned with each link are obtained by rotating $\hat{n}_x$:

$$\hat{a}_x = R_z(\theta_1)\hat{n}_x, \qquad \hat{b}_x = R_z(\theta_1 + \theta_2)\hat{n}_x, \qquad \hat{c}_x = R_z(\theta_1 + \theta_2 + \theta_3)\hat{n}_x, \tag{8}$$

$$\hat{d}_x = R_z(\theta_4)\hat{n}_x, \qquad \hat{e}_x = R_z(\theta_4 + \theta_5)\hat{n}_x, \qquad \hat{f}_x = R_z(\theta_6)\hat{n}_x. \tag{9}$$

Each rigid link i is written as a vector

$$\vec{v}_0 = l_0\hat{n}_x,\ \vec{v}_1 = l_1\hat{a}_x,\ \vec{v}_2 = l_2\hat{b}_x,\ \vec{v}_3 = l_3\hat{c}_x,\ \vec{v}_4 = l_4\hat{d}_x,\ \vec{v}_5 = l_5\hat{e}_x,\ \vec{v}_6 = l_6\hat{f}_x. \tag{10}$$

Key joint positions are then built by summing these vectors:

$$P_0 = O + \vec{v}_0, \tag{11}$$
$$P_1 = O + \vec{v}_1, \tag{12}$$
$$P_2 = P_1 + \vec{v}_2, \tag{13}$$
$$P_3 = P_2 + \vec{v}_3, \tag{14}$$
$$P_4 = P_0 + \vec{v}_4, \tag{15}$$
$$P_5 = P_4 + \vec{v}_5, \tag{16}$$
$$P_6 = O + \tfrac{1}{2}\vec{v}_0, \tag{17}$$
$$P_7 = P_6 + \vec{v}_6. \tag{18}$$

Points P_0 and P_6 lie on the rear chassis, while P_3, P_5, and P_7 lie on the moving "front" side of the spine.

7.2 Loop-Closure Constraints and Example Configurations

For a valid configuration of the Sarrus-based spine, the two branches must meet and the central link must connect consistently. three constraint equations are enforced:

$$\mathbf{c}_1 = P_3 - P_5 = \mathbf{0}, \tag{19}$$
$$\mathbf{c}_2 = P_5 - P_7 - \tfrac{1}{2}\vec{v}_3 = \mathbf{0}, \tag{20}$$
$$c_3 = \|P_0 - O\| - l_0 = 0. \tag{21}$$

In the Python script, these constraints are combined into a single vector $\mathbf{c}(\boldsymbol{\theta})$ and the joint angles $\boldsymbol{\theta} = [\theta_1, \ldots, \theta_6]^T$ are adjusted until $\|\mathbf{c}\|$ is near zero. By prescribing a desired value for one of the angles (for example θ_3, which roughly controls the amount of spinal bend) and solving for the others, several representative shapes of the spine are obtained:

- a neutral configuration (approximately straight spine),
- a moderately bent configuration,
- an oppositely bent configuration.

For each solution, the script evaluates the points P_0–P_7 numerically and plots the linkage in the plane, showing how the front and rear chassis remain parallel while the spine bends.

7.3 Jacobian and Force / Velocity Relationships

To understand how forces and velocities map between the actuators and the spine endpoint, the position of the point P_7 is differentiated with respect to a set of actuated joints. In this work, the joints at the right branch are treated as actuated, with generalized coordinates

$$\boldsymbol{\theta}_a = \begin{bmatrix}\theta_4 & \theta_5 & \theta_6\end{bmatrix}^\mathsf{T}. \tag{22}$$

The planar Jacobian is

$$J(\boldsymbol{\theta}) = \frac{\partial}{\partial \boldsymbol{\theta}_a}\begin{bmatrix}x_7(\boldsymbol{\theta})\\ y_7(\boldsymbol{\theta})\end{bmatrix}, \tag{23}$$

where $[x_7, y_7]^T$ are the x and y coordinates of P_7.

Evaluating J at a neutral configuration gives a numerical 2×3 matrix. This Jacobian is used in two standard ways:

1. **Force mapping.** A planar force applied at P_7, $\mathbf{F}_{\text{ee}} = [F_x, F_y]^T$, produces joint torques
$$\boldsymbol{\tau}_a = J^\mathsf{T}\mathbf{F}_{\text{ee}}. \tag{24}$$
Using a representative downward load of $F_y \approx 4\,\text{N}$ (comparable to the estimated peak load transmitted through the spine), the resulting torques at the actuated joints are on the order of $1\,\text{N m}$, consistent with the target torsional stiffness estimated in Sec. 6.

2. **Velocity mapping.** A desired velocity of the front chassis, $\dot{P}_7 = [\dot{x}_7, \dot{y}_7]^T$, is related to joint rates by
$$\dot{P}_7 = J(\boldsymbol{\theta})\,\dot{\boldsymbol{\theta}}_a. \tag{25}$$
For example, a small vertical motion $\dot{y}_7 \approx 0.05\,\text{m/s}$ at the neutral configuration requires joint speeds on the order of 0.1–0.2 rad/s, which are easily achievable by small geared motors.

This kinematic model confirms that the chosen spine geometry can satisfy the loop-closure constraints, that the resulting joint torques are compatible with the passive stiffness and expected actuators, and that reasonable endpoint velocities can be achieved with modest joint speeds.

8 Conclusion

We have designed, fabricated, and modeled a laminate articulated spine for a small quadruped robot inspired by both the directional stiffness of mammalian spines and the linkage-based backbones of myriapod-style robots. The mechanism uses a Sarrus-inspired parallel linkage implemented in a five-layer laminate stack, yielding stiff links and passive spring-like joints. A paper model and a laminate prototype confirm that the mechanism can bend as desired while maintaining vertical support. A Python-based kinematic model of the planar 3R–2R–3R parallel spine provides estimates of joint torques under expected ground reaction forces and joint velocities for prescribed body motions. These results establish the articulated spine as a feasible and tunable subsystem for future Jansen-leg quadruped prototypes.

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